Power Amplifier UI Prof Gunawan Wibisono Notes

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Reported by : Aziz Amerul Faozi

PCB stackup JLCPCB

Controlled Impedance PCB Parameters and Stackup

1.Prepreg dielectric constant:

Prepreg type	Dielectric constant
7628	4.4
3313	4.1
1080	3.91
2116	4.16

2.Solder mask Parameters

Coating Above Substrate C1	Coating Above Trace C2	Coating Between Traces C3	Coating Dielectric CEr
1.2mil	0.6mil	1.2mil	3.8

3. Core dielectric constant

Core dielectric constant
4.6

For your convenience, we have designed an Impedance Calculator to help you calculate the impedance and the trace width you require.

4-Layer Impedance Control Stackup

Thickness: 0.8mm | 1.0mm | 1.2mm | 1.6mm | 2.0mm

Outer Copper Weight: 1oz | 2oz

inner Copper Weight: 0.5oz | 1oz | 2oz

1) No requirement Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	
Core	Core	1.065mm	1.1mm H/HOZ with copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.035mm	

Figure 1: PCB Stackup

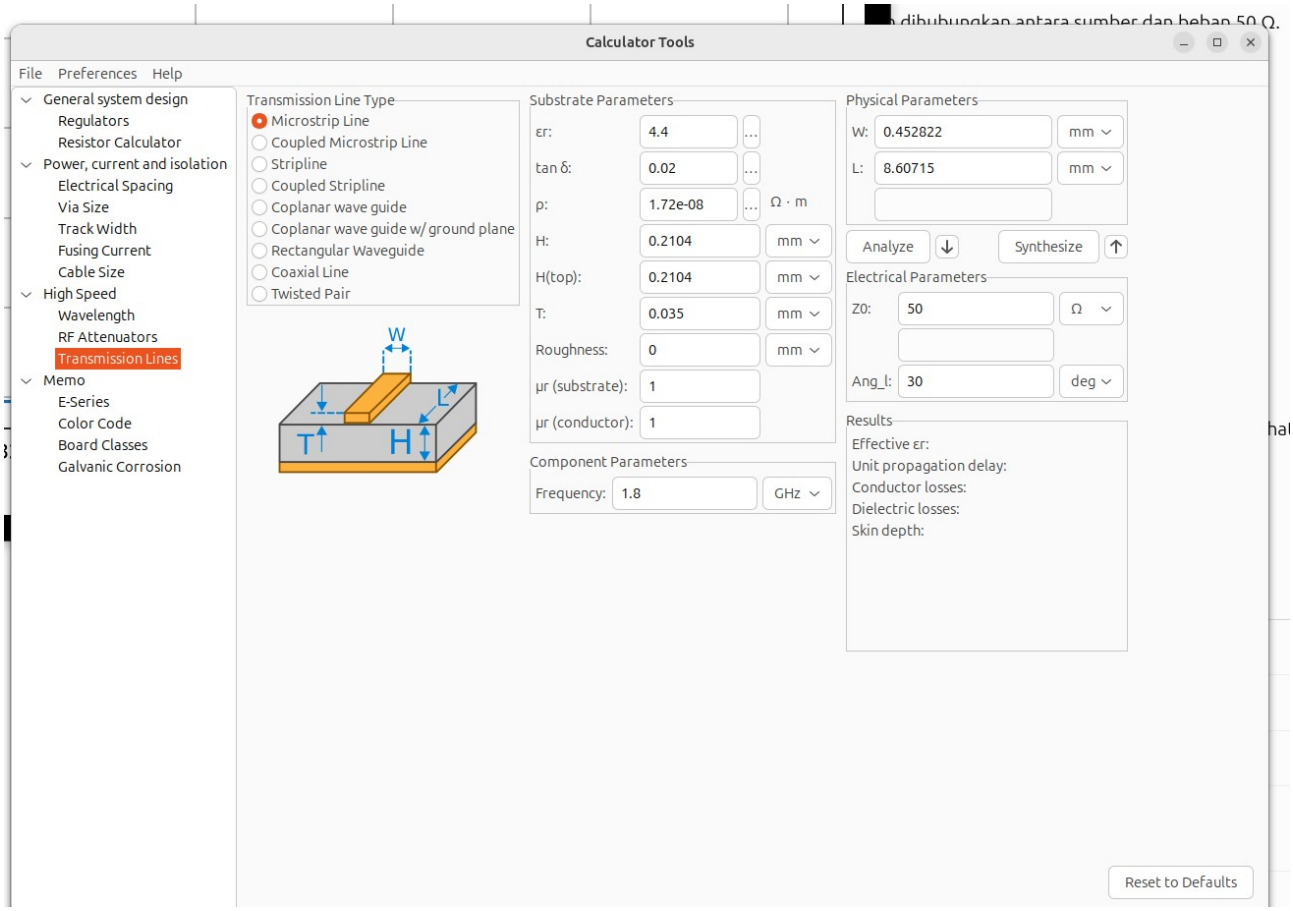


Figure 2: Kalkulator Ideal untuk 50 Ohm

Ketemu lebar jalurnya 4.52882 dengan pcb stackup dari jlcpcb supaya z0 50 ohm

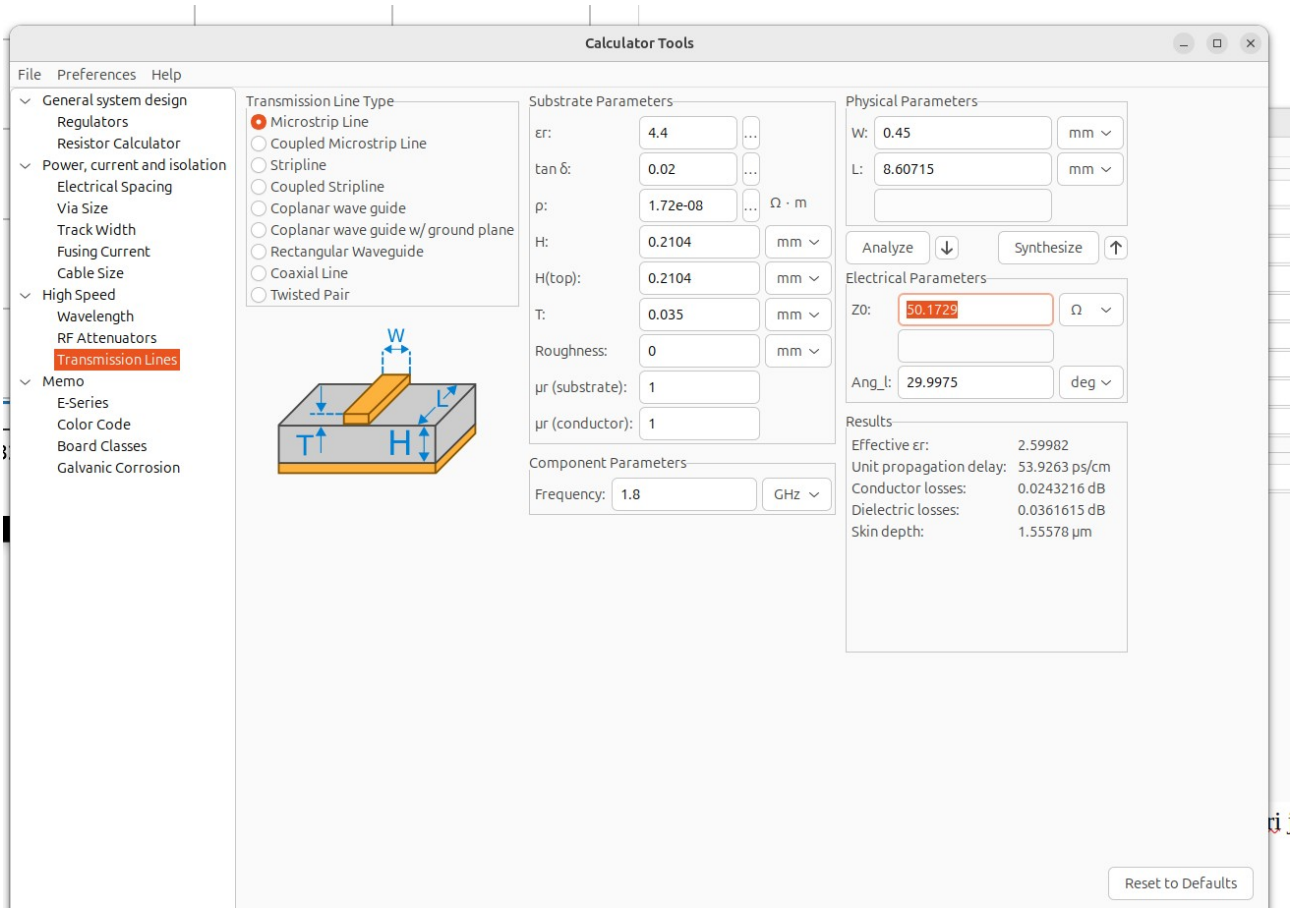


Figure 3: Calculator untuk feasible development

Pcb stackup optimized Z0 nya 50.1729 itung itungan vswr : 1.0036

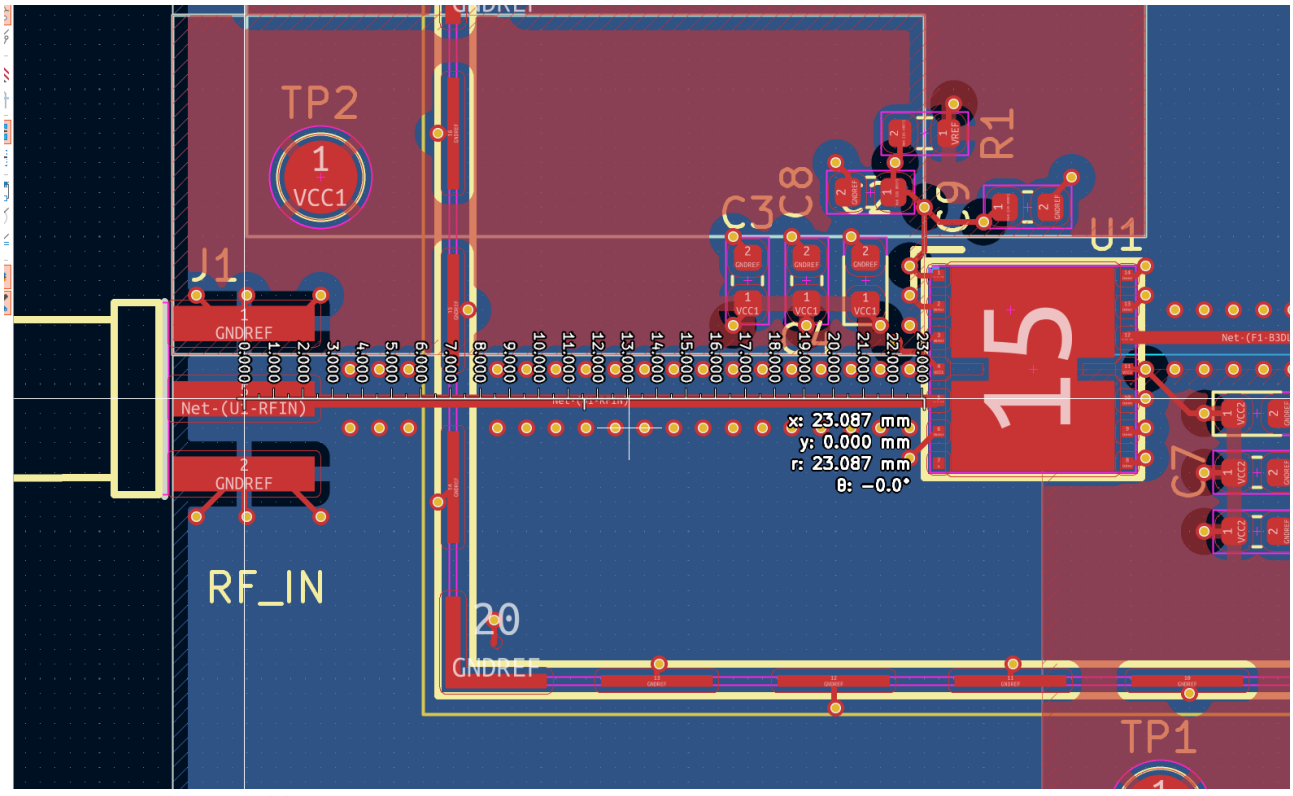


Figure 4: Jalur Feed ke PA

23 mm antara feed dan PA

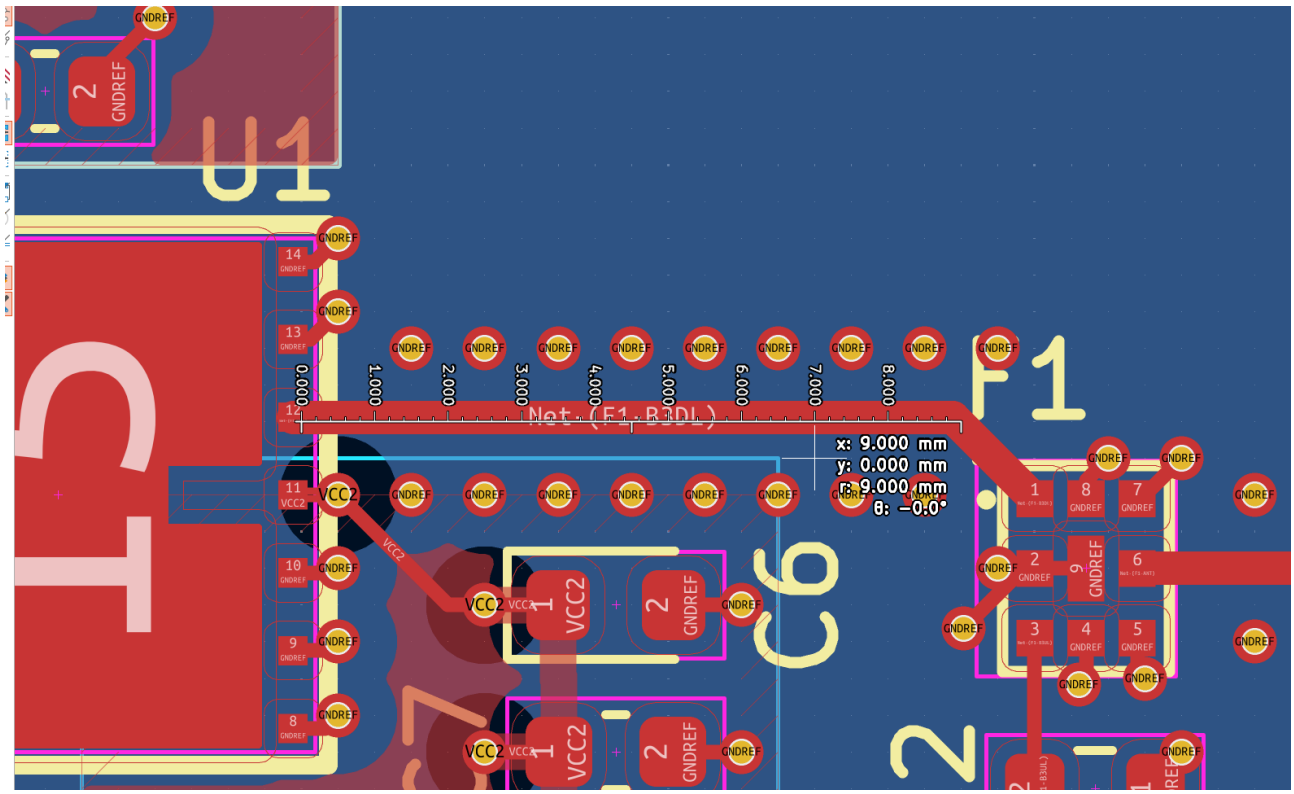


Figure 5: Jalur PA ke tikungan filter

9 mm antara output pa dan tikungan

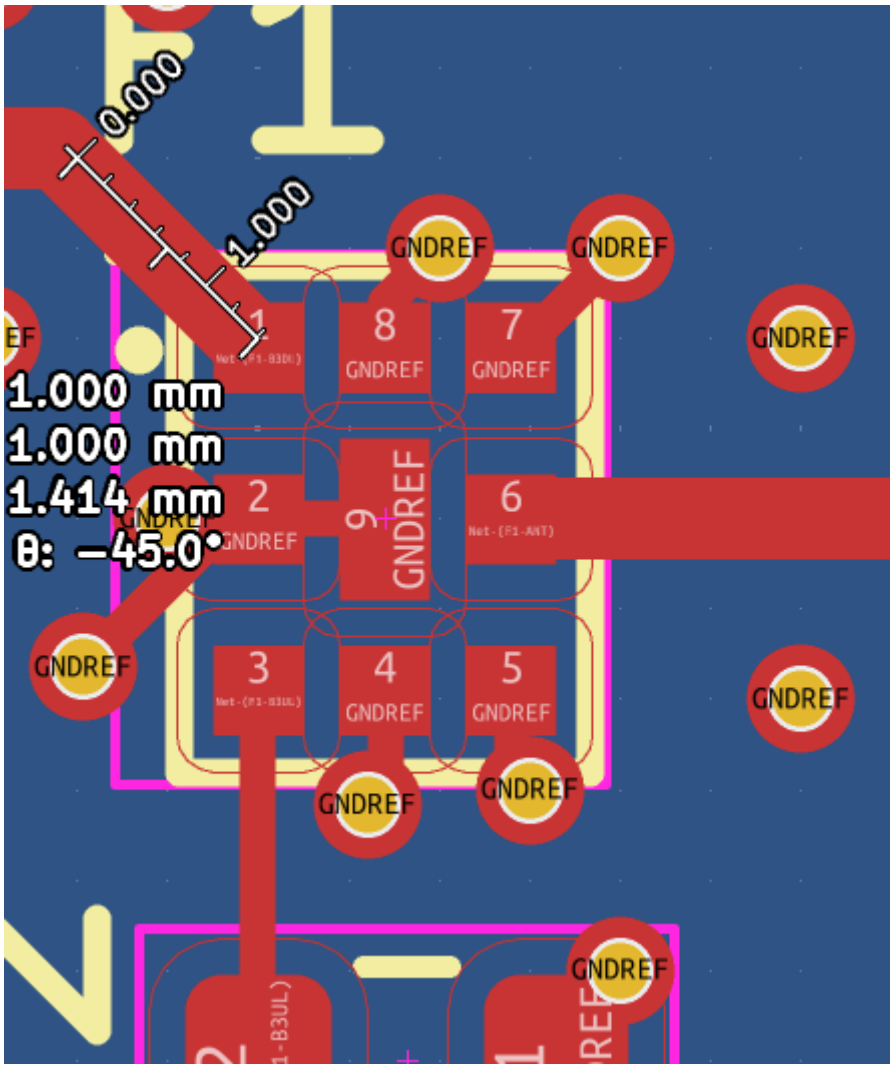


Figure 6: Jalur tikungan ke filter

tikungan dan filter 1 mm

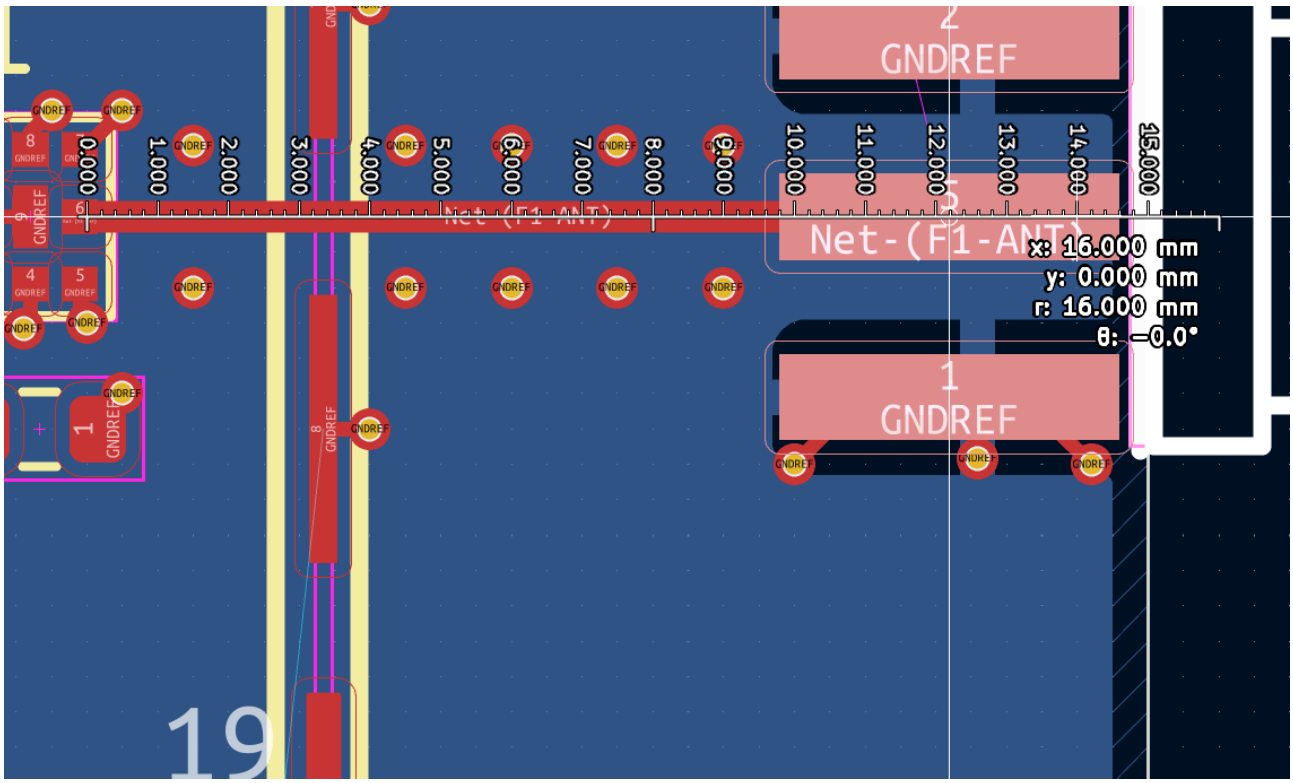
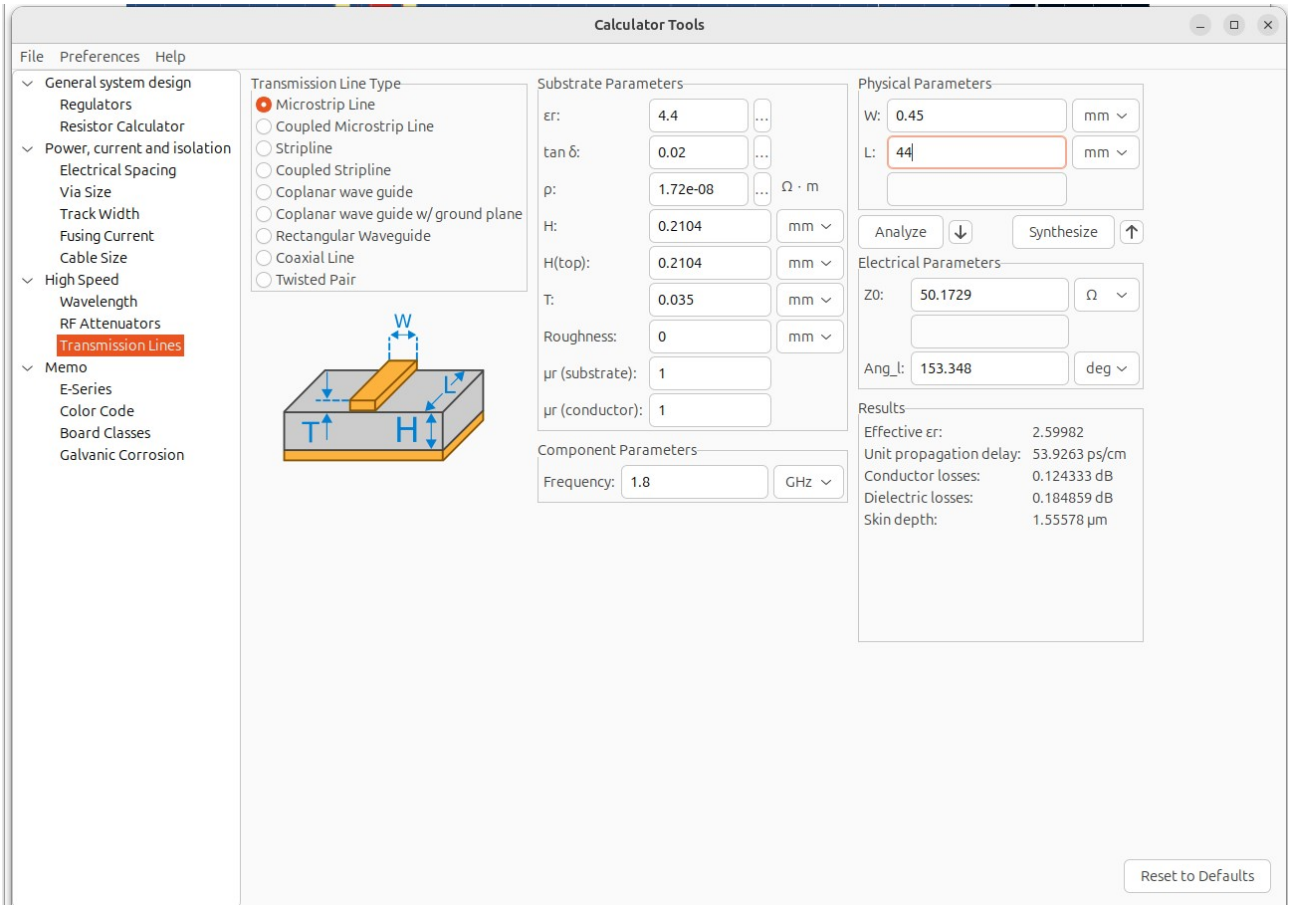


Figure 7: Jalur Filter ke Feed

14 mm filter ke feed

total jalur ditempuh 22+10+14 = 46 mm



total conduction loss

unit propagation delay : 53.92,63 ps/cm

conductor loses : 0.12433 dB

dielektrik lose : 0.184859 dB

perubahan fasa : 153.348

vswr PCB = (1+|(50.1729 - 50)/(50.1729 - 50)|)/(1-|(50.1729 - 50)/(50.1729 – 50)|)=1.0036

notes : untuk supply DC ataupun penyelelaras gelombang (siapa tahu ingin 0 degree) perlu requirement.